Set Up a Web Application on Amazon EC2

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1. Objective

To launch a virtual server on Amazon EC2, secure it with a Security Group that allows SSH (port 22) only from my IP and HTTP (port 80) for public access, connect to it over SSH, install the Nginx web server, deploy a simple HTML web application, and verify it in a web browser.

2. Architecture

User Browser → Internet → Security Group (ports 22, 80) → EC2 Instance (Ubuntu) → Nginx → index.html

Component	Role
EC2 instance	Virtual server hosting the application
Key pair (.pem)	Private key for secure SSH authentication
Security Group	Virtual firewall controlling inbound and outbound traffic
Nginx	Web server listening on port 80
index.html	The deployed web application

3. Step 1: Launch an EC2 Instance

1. Signed in to the AWS Management Console, opened EC2, and clicked Launch instance.
2. Name: my-web-server
3. AMI: Ubuntu Server 24.04 LTS (Free tier eligible)
4. Instance type: t2.micro / t3.micro
5. Key pair: created my-ec2-key (RSA, .pem format) and stored it safely.
6. Storage: default 8 GB gp3.
7. Clicked Launch instance.

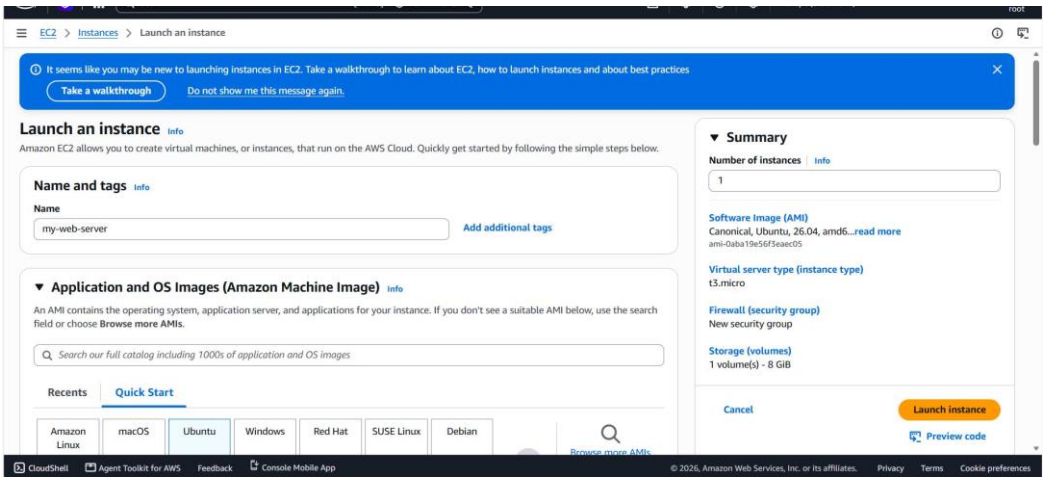


Figure 1: Launch instance screen in the EC2 console

Step 2: Configure the Security Group

A security group named web-server-sg was configured with these inbound rules:

Type	Protocol	Port	Source	Purpose
SSH	TCP	22	My IP	Admin access only from my machine
HTTP	TCP	80	0.0.0.0/0	Public web access

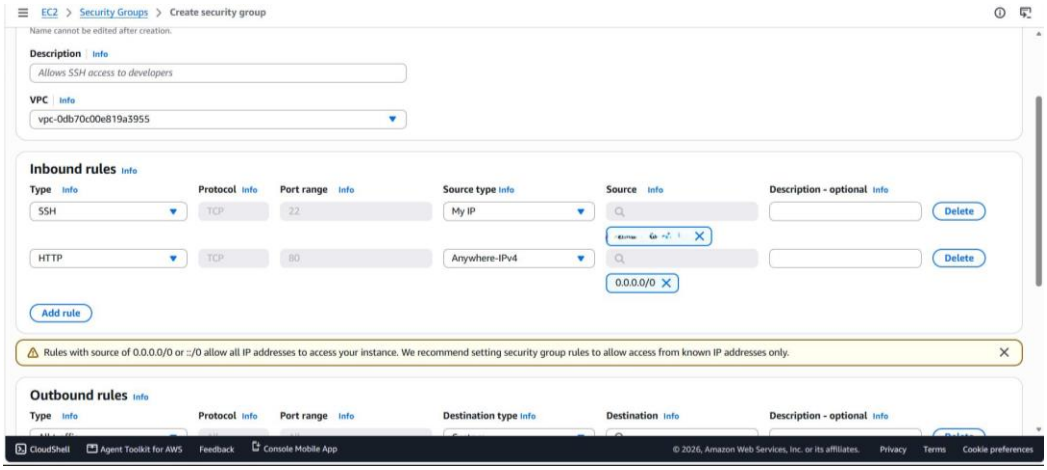


Figure 2: Inbound rules in the Create security group screen

Step 3: Connect to the Instance via SSH

Command run from Windows PowerShell:

```
ssh -i "C:\Users\Asus\Downloads\my-ec2-key.pem" ubuntu@ip
```

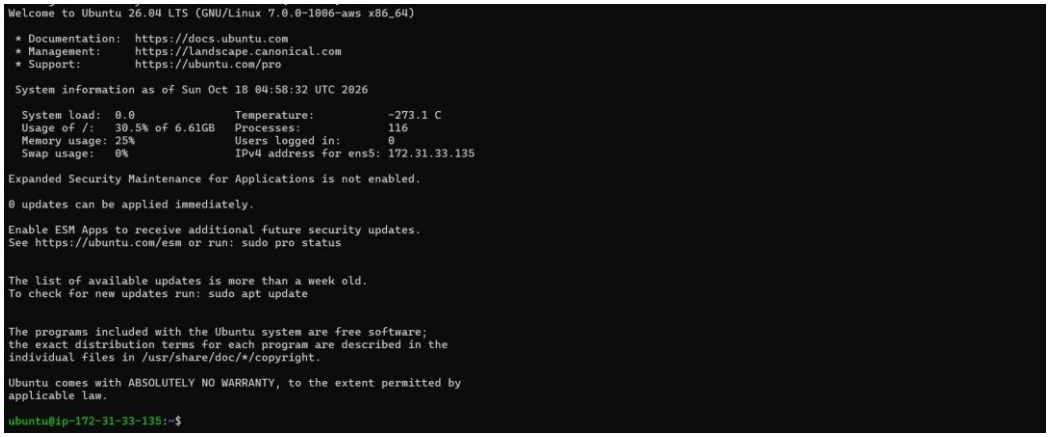


Figure 3: Successful SSH login to the Ubuntu instance

Step 4: Install and Configure Nginx

```
sudo apt update && sudo apt upgrade -y
sudo apt install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
sudo systemctl status nginx
```

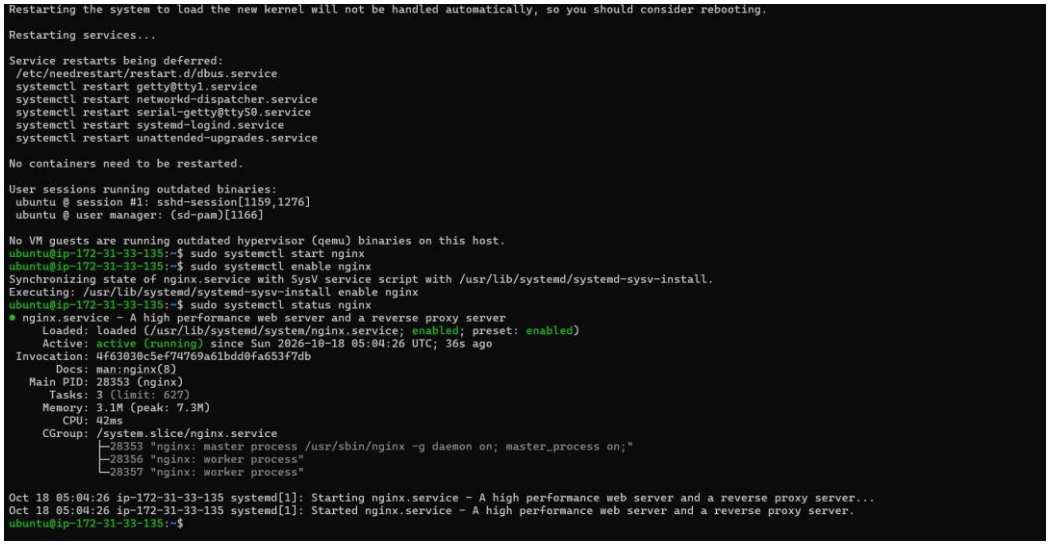


Figure 4: Nginx service running (systemctl status nginx)

Step 5: Deploy a Simple HTML Web Application

```
cd /var/www/html
sudo mv index.nginx-debian.html index.nginx-debian.html.bak
sudo nano index.html
```

Contents of index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My EC2 Web App</title>
<style>
body { font-family: Arial, sans-serif; background: #f0f4f8; text-align: center;
padding-top: 80px; }
h1 { color: #232f3e; }
p { color: #555; }
.badge { display: inline-block; background: #ff9900; color: #fff; padding: 8px 16px;
border-radius: 6px; }
</style>
</head>
<body>
<h1>Hello from Amazon EC2!</h1>
<p>This web application is hosted on an Ubuntu EC2 instance using Nginx.</p>
<span class="badge">Deployed by: Dhanashree Suresh Madhavai</span>
</body>
</html>
```

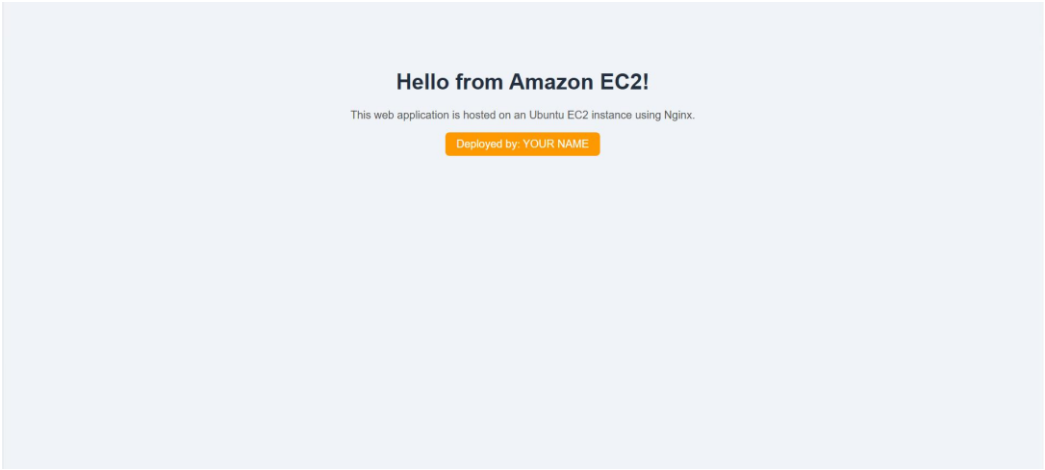


Figure 5: The deployed web application in the browser

Conclusion

An EC2 instance was launched, secured with a least-privilege Security Group, accessed over key-based SSH, configured with the Nginx web server, and used to host a simple HTML application reachable from a browser. The exercise covered virtual servers, network security, remote administration and web deployment, along with practical troubleshooting of connectivity problems.